A neutral parameter sweep: no effective size reproduces the empirical differentiation at the ancestry-tract-matched time

Module E working note — July 31, 2026

Abstract

We ran the recombination-matched, haplodiploid neutral null as a parameter sweep in carrying capacity K and founding size, sampling each of 20 independent demes every 25 ticks to 250. Two results follow. (1) **Effective size** scales with carrying capacity as $N_e \approx 0.31\,K$ (haplodiploidy and the age-structured life table pull N_e well below census K); founding size (100–200) barely affects N_e. (2) Using within-deme adjacent-marker LD as an ancestry-tract clock, the empirical value is matched by *every* parameter combination at only ∼15–20 generations of recombination. **At that tract-matched time the neutral among-population F_{ST} is ∼0.03–0.04, against the empirical 0.30 — a 6–10× gap — and no combination, at any time out to 74 generations including the strongest drift, comes near it.** Insufficient recombination is excluded, and no neutral effective size in this range rescues the differentiation. This is the parameter-swept, direct form of the F_{ST}–tract dissociation.

1 The sweep

Base configuration $K = 6250$, total founders = 150 (drawn from the diversified 1200/520 pool, split aquilonia-males : polyctena-queens to hold admixture proportion $\alpha \approx 0.54$). One-at-a-time: $K \in \{3125, 6250, 12500\}$ and total founders $\in \{100, 150, 200\}$, others at base — five combinations, 20 demes each, one replicate. Every deme runs to 250 ticks and is sampled every 25. Markers are the **9,902 high-DiagnosticIndex loci** (DI > -25, MAF ≥ 0.15, systematically thinned to ∼15 kb spacing), the ancestry-informative set that tract-length estimation requires. **Ticks are not generations**: the life table gives a generation time of ≈ 3.4 ticks, so 250 ticks ≈ 74 generations; all generation axes below are ticks/3.4.

2 Effective size (Fig. 1A, Table 1)

Within-deme expected heterozygosity decays log-linearly with generation, giving N_e from the slope ($H_e(g) = H_e(0)\,e^{-g/2N_e}$).

Table 1: Effective size from within-deme heterozygosity decay (generations = tick/3.4).

combination	K	founders	N_e	fit R^2
K3125_f150	3125	150	1,195	0.99
K6250_f150	6250	150	1,950	0.99
K12500_f150	12500	150	3,917	0.93
K6250_f100	6250	100	1,980	0.99
K6250_f200	6250	200	2,105	0.99

$N_e \approx 0.31\,K$ across the K sweep. Founding size (100–200) changes N_e negligibly: the bottleneck sets the *initial* differentiation, not the long-run drift rate. So K is the effective-size lever and founding size is a starting-condition lever. Note that every simulated deme retains $H_e \approx 0.47$, whereas the empirical value on the same markers is $H_e = 0.31$ (dashed) — the empirical diagnostic loci are far more sorted (diversity-depleted) than neutral drift produces.

3 The tract-matched dissociation (Fig. 1B,C)

Panel C uses within-deme adjacent-marker r^2 as an ancestry-tract clock: it is the admixture LD that recombination erodes as tracts shorten. The empirical value (0.556) is crossed by *all five* combinations at ~15–20 generations (ticks ~50–75); the curves overlie, so the tract clock is nearly parameter-independent and reads ~15–20 generations — consistent with the ~20-generation field age of the hybrid populations.

Panel B is the differentiation at those same times. At the tract-matched ~15–20 generations the simulated among-deme F_{ST} is ~0.03–0.04, against the empirical **0.30**. The gap is 6–10×, and it does not close anywhere in the sweep: even the strongest drift (K3125, $N_e = 1195$) at 74 generations reaches only $F_{ST} = 0.055$, by which time its tracts (adjacent LD 0.47) have already fallen *below* empirical. For drift alone to reach $F_{ST} = 0.30$ in ~20 generations would require $N_e \approx 30$ ($K \approx 100$) — implausible, and it would deplete diversity genome-wide rather than concentrate differentiation at diagnostic loci.

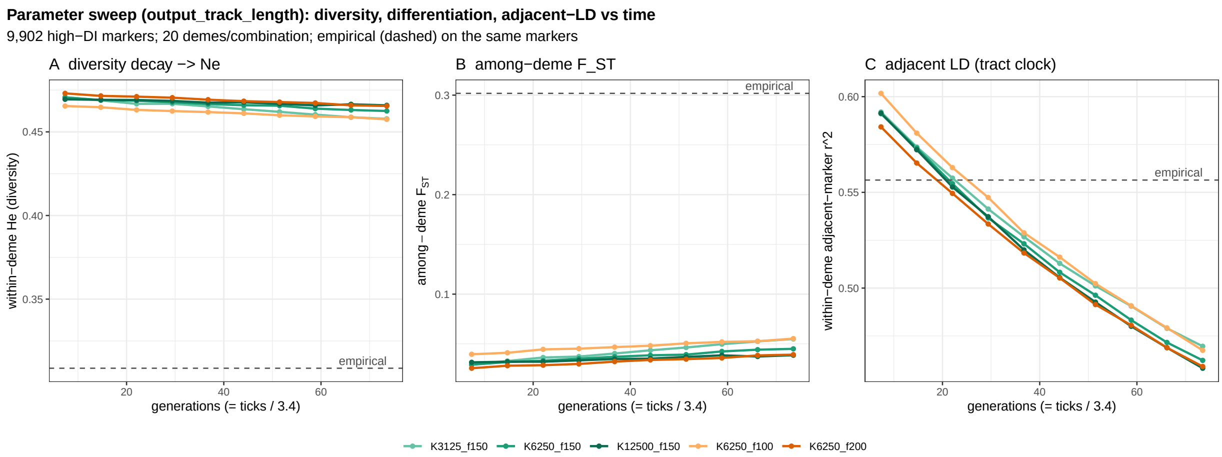

Figure 1: Parameter sweep vs generations (= ticks/3.4); empirical on the same 9,902 high-DI markers is dashed. **(A)** within-deme diversity decay (source of N_e); all sims sit far above the empirical (heavily sorted) value. **(B)** among-deme F_{ST} — every combination is ~0.03–0.05, an order of magnitude below empirical 0.30. **(C)** within-deme adjacent-marker LD (tract clock) — the empirical value is matched at ~15–20 generations. Reading B at the C-matched time gives the 6–10× F_{ST} deficit.

4 Conclusion

Across a $K\times$ founding-size grid spanning $N_e \approx 1200$–3900, no neutral combination reproduces the empirical among-population differentiation at diagnostic loci at the time its ancestry tracts match empirical — nor at any time in the 74-generation window. The differentiation deficit is 6–10× and parameter-robust, and the diversity is simultaneously far too high. This is the direct, parameter-swept statement of the result the earlier analyses inferred: **the empirical high-F_{ST}, tract-appropriate, diversity-depleted state at ancestry-informative loci is not a neutral outcome at any effective size or age in this range; it is the signature of locus-specific ancestry sorting.** Insufficient recombination, insufficient drift, and founding-bottleneck strength are all excluded as explanations.

Preliminary in one respect. Panel C uses within-deme adjacent-marker LD as a *proxy* for tract length. The colleagues' ancestry-tract caller will give the actual tract-length distribution (in cM), pinning the matched generation more precisely and allowing the full distribution shape to be matched rather than a single LD summary. The F_{ST}–tract dissociation is already unambiguous and does not depend on the proxy.

Reproducibility

Sweep harness and inputs: `formica_neutral_sim_sweep_handoff/` (README, `run_sweep.sh`, model, `markers.tsv`, founders). Output: `moduleE_slim/output_track_length/` (5 combinations $\times$ 20 demes $\times$ 10 ticks). Analysis: `dev/R/moduleE_analyze_sweep.R` (Fig. 1, Table 1). Generation time from the life table; $\sim$3.4 ticks/generation.